

2-Axis Ground Station & 5.8 GHz Dish

Primary designer / CU Sounding Rocket Laboratory

I designed the gimbal and integrated dish system, carried the mechanical design through fabrication and assembly, and now direct new members developing harnessing and PCB fixtures for the elevation stage.



SRL team with the assembled ground station and dish structure

~0.2 degrees average angular error

Commanded versus measured angle across a small set of physical tests.

Mechanical architecture

The system combines an azimuth turntable with a driven elevation stage, dish attachment structure, and counterweights. The design review specifies 180 degrees elevation travel and a maximum 10 lb antenna load. My design work includes the drivetrain, structural hardware, dish structure, and antenna fixture.

Sizing and structural checks

The design review includes hand calculations for shaft torsion and motor-standoff bending, alongside ANSYS static structural analysis of an acceleration load case. The ANSYS deformation plot reports a maximum of 1.3774×10^{-5} m, approximately 0.014 mm. The review reports 9.687 MPa maximum stress for that case. These are model predictions under the review assumptions; the review identifies non-constant acceleration effects as further work.

Drivetrain and fabrication

The motor, coupler, shaft, and supporting components form the elevation drivetrain. The parts list combines machined parts, waterjet-cut and bent sheet metal, printed counterweight housings, and purchased bearings and fasteners. Tolerance analysis and assembly were part of my design work.

Assembly and integration

I worked with the SRL team to manufacture and assemble the system. Current integration work includes directing members on harness routing and PCB fixtures for the elevation section. I also teach newer members CAD modeling and mechanical design techniques.

Measured positioning performance

We compared commanded angles with physically measured angles over a small set of tests and obtained approximately 0.2 degrees average error. This is an average positioning result, not a maximum-error bound or a dynamic tracking result. The review's 0.009 degrees microstep increment describes command resolution and is a different quantity from measured error.

Integrated dish

I was the primary designer of the integrated 5.8 GHz dish structure and antenna fixture. The dish belongs to the same ground-station project, alongside the gimbal and its mechanical interfaces.

Sources: supplied design review or research presentation, with project ownership, test method, and current status clarified by Landon Holligan. Original PowerPoint files accompany the website.